



MASTG

Algorithms

- MASTG-TEST-0015: Testing the Purposes of Keys
- MASTG-TEST-0016: Testing Random Number Generation
- MASTG-TEST-0204: Insecure Random API Usage
- MASTG-TEST-0205: Non-random Sources Usage
- MASTG-TEST-0208: Insufficient Key Sizes
- MASTG-TEST-0212: Use of Hardcoded Cryptographic Keys in Code
- MASTG-TEST-0221: Broken Symmetric Encryption Algorithms

L1

L2

ANDROID

MASWE-0027

TEST



MASTG-TEST-0205: Non-random Sources Usage

Overview

Android applications sometimes use non-random sources to generate "random" values, leading to potential security vulnerabilities. Common practices include relying on the current time, such as `Date().getTime()` , or accessing `Calendar.MILLISECOND` to produce values that are easily guessable and reproducible.

Steps

1. Run a static analysis ([🔗 Static Analysis on Android](#)) tool on the app and look for uses of non-random sources.

Observation

The output should contain a list of locations where non-random sources are used.

Evaluation

The test case fails if you can find security-relevant values, such as passwords or tokens, generated using non-random sources.

Best Practices

🤖 MASTG-BEST-0001: Use Secure Random Number Generator APIs

Demos

🤖 MASTG-DEMO-0008: Uses of Non-random Sources

